

Seh Na Mellick

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Machine learning, automated microscopy, and evolutionary modeling for understanding and treating bacteria.

EDUCATION

Carnegie Mellon University Pittsburgh, PA, 2024 – Present

Ph.D. Computational Biology, GPA: 3.8 (expected 2029)

- Coursework: Machine Learning (PyTorch) · Computational Structural Biology · Cellular Systems Modeling · Computational Genomics · Modeling Evolution · Advanced Genetics
- Teaching: 02-712 Computational Methods for Biological Modeling and Simulation (2024) · 02-250 Introduction to Computational Biology (2025)

University of Nebraska Omaha Omaha, NE, 2019 – 2024

B.S. Molecular & Biomedical Biology, GPA: 3.9, Minors: Chemistry & Mathematics

- Graduate Coursework: Discrete Event Simulations Modeling · Boolean Network Models
- Undergraduate Coursework: Molecular Genetics · Cellular Biology · Immunology · Molecular Neurobiology · Organic Chemistry · Biochemistry · Machine Learning & Data Mining · Multivariable Calculus · Linear Algebra · Probability & Statistics · Bioinformatics · Deterministic Operations Research · Discrete Mathematics

Central High School Omaha, NE, 2016 – 2019

GPA: 4.8, AP Scholar

RESEARCH EXPERIENCE

Graduate Research Assistant Carja and Bridges Labs @ Carnegie Mellon University, 2024 – Present

- Leveraged automated high-throughput image-based phenotyping to generate large-scale datasets characterizing growth phenotypes of bacterial gene deletion mutant libraries under various conditions.
- Designed image analysis and computer vision pipelines combining vision transformers, quantitative feature extraction, and representation learning to characterize spatiotemporal bacterial growth phenotypes.
- Built models to predict bacterial growth dynamics, classify gene deletions and species, predict antibiotic resistance phenotypes, and identify novel genotype-phenotype relationships.
- Modeled the influence of collective aggregation on evolution of antibiotic resistance in bacterial infections using theoretical population genetics, evolutionary rescue modeling, and stochastic simulations.

AI PhD Resident Google X, The Moonshot Factory, 2026

- Conducted research on early-stage investigation involving AI and LLMs for biological lab automation, building agentic workflows with LangGraph.
- Developed skills in computer & machine vision, automated experimentation, woodworking, metalworking, and welding.

Rotation Student Cooper Lab @ Univ. of Pittsburgh, 2024

- Developed experimental and computational models of catheter-associated bacterial infections and performed evolution experiments to validate model predictions of antibiotic resistance evolution *in vitro*.
- Prepared and analyzed Illumina whole-genome sequencing libraries for barcoded samples.

Microbiology Consultant UNeTECH @ Univ. of Nebraska Medical Center, 2021 – 2022

- Designed and conducted microbiology experiments for local businesses and healthcare partners.
- Analyzed and communicated experimental results to industry partners and non-technical stakeholders.

Undergraduate Research Assistant Rowen Lab @ Univ. of Nebraska Omaha, 2020 – 2024

- Performed molecular cloning and microbiology experiments to characterize the role of the putative transcription factor PA5189 in *Pseudomonas aeruginosa* antibiotic resistance and biofilm formation.
- Analyzed RNA-seq and whole-genome sequencing data to identify mutations in strains expressing resistance to a novel antimicrobial peptide.

Research Assistant Soukup Lab @ Creighton University, 2018 – 2019

- Performed molecular cloning and microbiology experiments to assess *glmS* riboswitch ligand analogs for potential use as antibacterial agents in *Staphylococcus aureus* and *Bacillus subtilis*.
- Generated and prepared crystals for X-ray crystallography to determine mRNA riboswitch 3D structures.

PRESENTATIONS & PUBLICATIONS

1. **Mellick, S.**, Rowen, D. (2024). Characterizing the role of PA5189 in *Pseudomonas aeruginosa* deletion and overexpression mutants. *Undergraduate Honors Thesis*, University of Nebraska Omaha.

2. **Mellick, S.**, Rowen, D. (2023). Role of the PA5189 regulon in *Pseudomonas aeruginosa* antibiotic resistance and biofilm formation. *Conference talk*, Nebraska Academy of Sciences, Lincoln, NE.

SELECTED COURSE PROJECTS

Predicting Fitness Effects of Synonymous Mutations 10-710 *Computational Genomics*, CMU, 2025

Curated a dataset integrating genomic, evolutionary, kinetic, and structural features of synonymous mutations in *S. cerevisiae* and built interpretable models predicting fitness effects of synonymous mutations.

Probing Chirality in E(3)-Equivariant Boltzmann Generators MSCBIO 2030 *Comp. Structural Biology*, Pitt, 2025

Demonstrated that SO(3)-equivariant Boltzmann generators are necessary to capture chirality in molecular structures and latent representations, with implications for using traditional E(3)-equivariant models for modeling biomolecular systems and drug design.

Adversarial Robustness in ImageNet 10-701 *Machine Learning*, CMU, 2024

Studied transferability of localized adversarial perturbations across CNN and Vision Transformer architectures using FGSM, PGD, and Carlini-Wagner adversarial techniques combined with segmentation and bounding-box detection to localize perturbations to specific image regions.

AWARDS & HONORS

Research Awards: Best Poster, CMU Computational Biology Symposium (2026) · 2023 UNO FUSE Grant · 2022 NASA Nebraska Space Grant Fellowship.

Academic Honors: UNO Honors Distinguished Scholar · Omaha Hearing School Scholarship (academic achievement as Deaf/Hard-of-Hearing student) · UNO Chancellor's Scholarship

Music Performance: Nebraska Wind Symphony Memorial Concerto Competition Scholarship (2021) · recipient of five merit-based scholarships for music performance at UNO School of Music

SKILLS

Machine Learning & AI: PyTorch · scikit-learn · vision transformers · CNNs · LLM systems & fine-tuning · LLM agents (LangGraph)

Programming: Python · Julia · R · Git · Bash · Docker/Singularity · HTML, CSS, & JS

Concepts: statistical learning · active learning · HPC · networked systems · computer & machine vision · deterministic & stochastic modeling

Biology: microbial cloning & molecular genetics (PCR, qRT-PCR, plasmid construction, transformation, conjugation) · experimental microbiology (MIC & biofilm assays, microbial imaging) · experimental evolution · mammalian cell culture · bioinformatics (WGS, bulk RNA-Seq, barcode sequencing, phylogenetics) · automated microscopy · liquid handling robotics · brightfield & confocal microscopy image analysis pipelines

Languages: English (native) · French (intermediate) · American Sign Language (novice)

MEMBERSHIPS & ACTIVITIES

Senator: CPCB Graduate Student Assembly (2024–Present)

Member: American Society for Microbiology, (2024–Present) · UNO Honors Program (2019–2024)

Bassist: CMU Indian Music Ensemble (2024–Present) · UNO Maverick Machine (2023–2024)

Principal Oboist: UNO Symphonic Wind Ensemble (2019–2024) · UNO Heartland Philharmonic Orchestra (2019–2023) · UNO University Band (2021–2023)